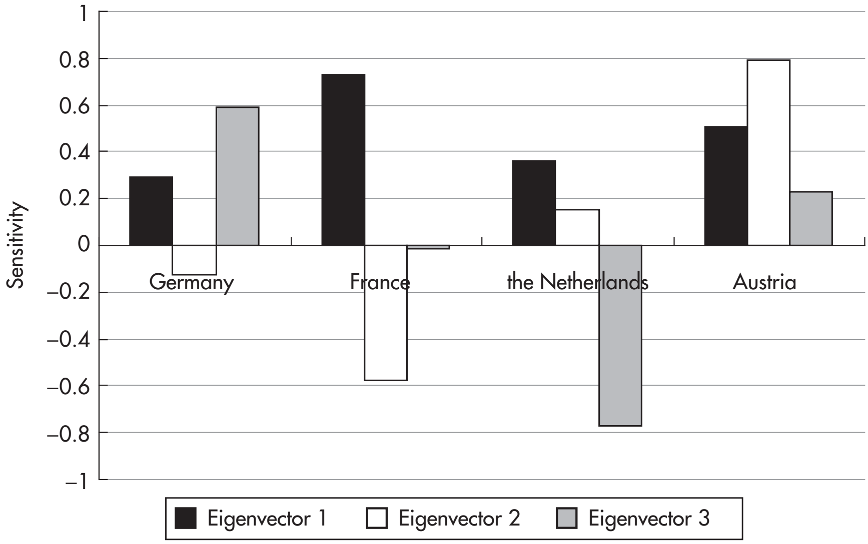

FIGURE 3.9 First three eigenvectors of a PCA on 5Y CDS for core Eurozone sovereign issuers.
Sources: data – Bloomberg; chart – Authors.

Data period: 6 May 2009 to 26 Sep 2012, weekly data.

than Germany. The second eigenvector groups together Germany and France (negative sensitivities) versus the Netherlands and Austria (positive sensitivities). Thus, if factor 2 increases, the CDS of the small core countries widen relative to the CDS of the big core countries. Therefore, factor 2 can be interpreted as differentiating between the big and small countries in the core Eurozone. This means that the size of the bond market is the second-most-important determining factor of core Eurozone countries' CDS levels (after the overall CDS level as measured by factor 1).

For an example from the commodity market we have run a PCA on weekly data for the front month contract on the three soy-related series (soybeans, soybean meal, and soybean oil) from 2000 onward. The scaled eigenvalues shown in Figure 3.10 indicate that almost everything is explained by factor 1, with factor 2 having only 0.2% of explanatory power and factor 3 virtually nothing. This indicates that differentiation across the three soy products is limited relative to the variability of the overall price changes in the three commodities.

Figure 3.11 displays the eigenvectors. Note that the difference in sensitivities to the first factor is a function of the difference in the size of the

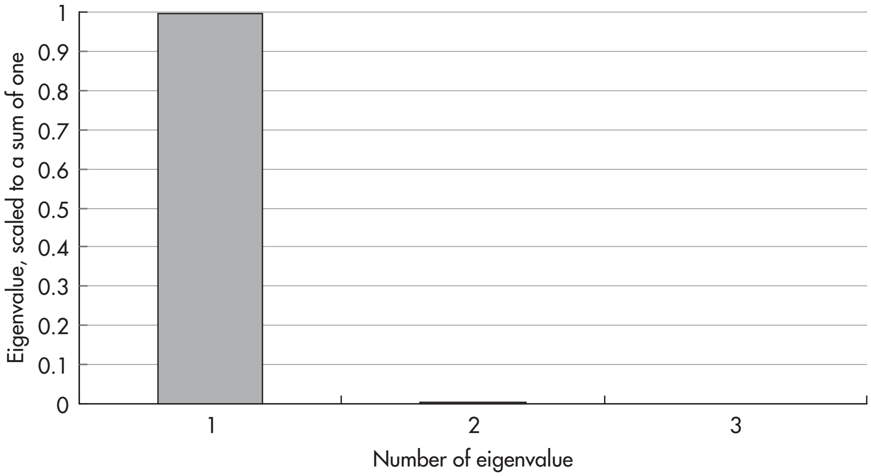

FIGURE 3.10 Scaled eigenvalues of a PCA on the soy market.
Sources: data – Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 6 Aug 2012, weekly data.

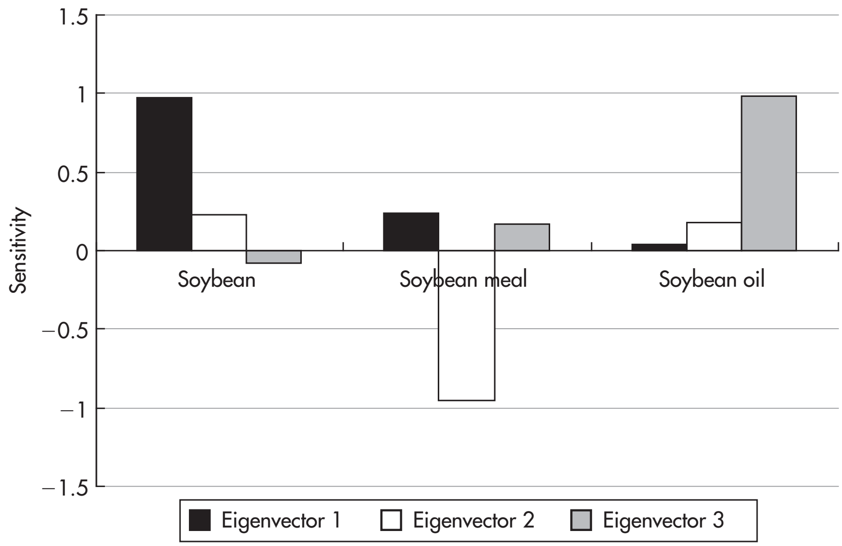

FIGURE 3.11 Eigenvectors of a PCA on the soy market.
Sources: data – Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 6 Aug 2012, weekly data.

input numbers (e.g. 1,600 for soybeans versus 51 for soybean meal). This could be avoided by creating synthetic time series with similar numbers, for example, all starting with a value of one. It turns out that the overwhelming factor 1 affects all soy products in the same way (i.e. all rise and fall together). The (little) differentiation between different soy products which is measured by factor 2 shows that soybeans and soybean oil move together versus soybean meal.

Finally, a look at the evolution of the PCA factors over time (Figure 3.12) reflects the increasing demand for soy products over the last decade in a rise of factor 1, which has led to all three products richening. Factor 2, on the other hand, has not exhibited a clear trend, which means that the price differentiation between soybeans and soybean oil, on the one hand, and soybean meal, on the other, tends to be temporary (i.e. factor 2 tends to be mean reverting). From this historical perspective, the current unprecedented deviation of factor 2 from its mean could be seen as a good trading opportunity. Factor 2 being too low (historically) translates through the sensitivities to eigenvector 2 in soybean meal being too expensive versus soybeans and soybean oil. Hence, a low factor 2 reflects the relative richness of soybean meal versus other soy products.

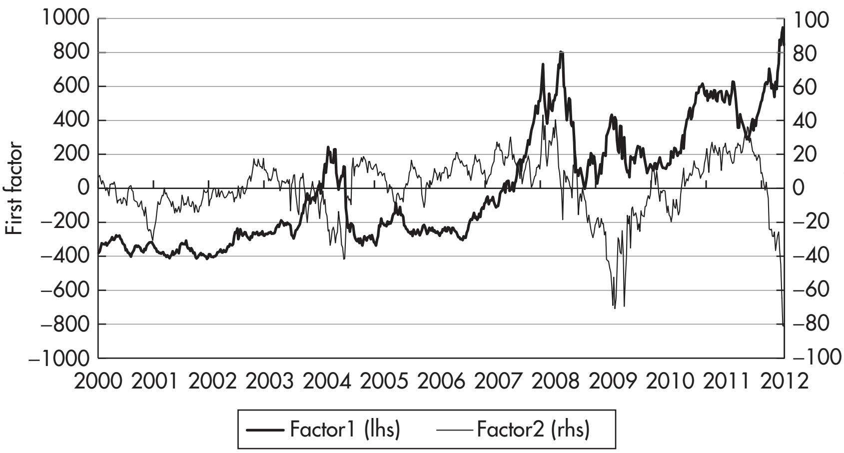

FIGURE 3.12 Historical evolution of the first and second factors of a PCA on the soy market.
Sources: data – Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 6 Aug 2012, weekly data.

Up until this point, the analysis has been purely statistical, revealing the driving forces of historical and current pricing. This provides a good basis for a trading decision, which needs to incorporate elements besides statistical properties as well. For example, is the current deviation of factor 2 an outlier that can be expected to revert to mean as quickly as it did in the past? Or is there a potential for the current drought to cause a permanent regime shift in the soy markets? While this assessment is beyond the reach of statistics, PCA both enables us to detect and formulate these trading decisions and indicates that any cause for a permanent regime shift would need to be extraordinarily strong – stronger than anything that has happened over the past decade. Thus, PCA shows that arguing for a regime change (i.e. for no reversion of factor 2 to the longstanding mean) would require very good reasons.

An investor believing in the current drought not to be such an extraordinary regime-changing event and thus in factor 2 continuing its decade-long behavior and returning to its mean may therefore want to consider selling soybean meal versus either soybeans or soybean oil. Which of the two is better? And precisely how much should be bought for one short soybean meal futures contract? The answer to these questions is again within the realm of statistics. Given the much stronger impact of factor 1 on the soy market, there is actually a high risk that a position that intends to exploit the current mismatch in factor 2 could end up being driven mainly by factor 1 and not 2 (i.e. overall direction of soy, not the spreads between different soy products), if hedge ratios are not calculated properly. Subsequently, we shall develop PCA into tools that answer these questions.

These examples could serve to give an impression of the analytical strength of PCA, thereby empirically proving our claim that PCA is a very useful tool for the relative value craftsman to dig out and clarify the key market mechanisms hidden below a muddy surface of incommensurable market actions.

So far, we have used PCA to gain insights into the structural core of markets. From now on, we shall focus on applying these insights to find, analyze, and construct trading ideas. This will take place in the framework of PCA and thus show the way PCA supports the translation of its insights into practical trading positions.

DECOMPOSING MARKETS INTO UNCORRELATED FACTORS

As the eigenvectors are orthogonal, the factors are uncorrelated by construction. Hence, one can decompose a complex market into individual, uncorrelated, and simple relationships. This ability is the key to a variety of analyses.

Technical Points

- Factors are uncorrelated, not necessarily independent. This issue has led to the development of independent component analysis recently, though we have not seen a convincing application to market analysis yet.
- While factors are uncorrelated over the whole sample period, there can occur significant measured correlation within subperiods. Thus, the rolling correlation of factors should be investigated before relying on the model. This is a key point, which we shall discuss in detail toward the end of this chapter.

The decomposition into uncorrelated factors isolates a particular relationship from others. This makes it possible to analyze and trade individual market mechanisms, such as yield curve steepness, without being influenced by other effects, such as direction.[8] The most important application of this ability is the exclusion of directional effects (typically associated with the factor with the greatest explanatory power). If relative value trades are defined as offering return opportunities that are uncorrelated to market direction, then the ability of PCA to produce and analyze time series uncorrelated with market direction is a key aspect of the analysis.[9] In other words, if the first factor represents the market direction (beta), then alpha (P&L uncorrelated to the market direction) can be found in the other factors.

Using the example of the Bund yield curve from Figure 3.5, the second factor has been interpreted as explaining the steepness not explained by the first, directional factor (i.e. representing the *non-directional steepness*). Note that also the first factor has a steepness component (i.e. direction impacts steepness). If yields increase, the 5Y-10Y yield curve tends to flatten. The second factor is uncorrelated to direction *and to directional impacts on the steepness* and therefore shows the steepness of the curve *net* of directional impacts.

To repeat this key point: given the current yield level, the curve should have a certain steepness, given by the first eigenvector. The second factor shows the steepness of the curve that remains after taking into account the

[8]This requires the assumption that the future curve dynamic will be the same as the current curve dynamic expressed in PCA eigenvectors. Toward the end of this chapter we shall investigate the validity of this assumption.

[9]Likewise, one could use the same techniques to analyze and execute trading positions, which are uncorrelated to, for example, non-directional steepness (represented here by the second factor). However, in practice, most of the time the main concern will be to prevent the all-pervasive directional effects from influencing relative value positions, in other words to create alpha.

steepness already explained by the direction. There were many instances in which the 5Y-10Y yield curve has been steep but in which the steepness was due entirely to the low yield level, while on a non-directional basis the curve was actually too flat. In this case, one could argue for a non-directional steepening trade hedged against directional impacts, despite the steepness of the 5Y-10Y curve.

This example illustrates the importance of factor decomposition for RV analysis:

- Given the all-pervasive directional effects (e.g. on curve steepness, curvature, swap spreads, volatility) the indispensable prerequisite for any RV analysis is a measure for steepness or curvature *unaffected* by directional effects. The second and third factors provide that measure and thus the *starting point* for any RV analysis.
- As the first eigenvector shows the way yield levels affect slope, we can hedge against these effects, thereby formulating RV trades with no directional exposure.
- PCA is thus a direct way to gain access to true RV trades and thereby to much more trading, sales, and research opportunities than straight directional positions. Simply put, by using a three-factor model, one obtains three uncorrelated time series and thus three times as many possibilities for trades.

EMBEDDING PCA IN TRADE IDEAS

So far, we have discovered two important features of PCA: its ability to decompose a market into uncorrelated factors and the possibility of interpreting these factors economically (via examining the shape of the eigenvectors).[10] Together, PCA decomposes a market into uncorrelated factors with an economic meaning.

The process of taking a view on the market therefore can be achieved by taking a view on each of the factors. Hence, for each factor, the analyst can decide independently whether to take a view on that factor. His decision should reflect:

- statistical criteria like mean reversion
- fundamental and structural criteria
- flow and other concerns.

[10] While the shape of the eigenvector always allows an economic interpretation of the factors such as "yield curve steepness," it may not always be possible to link each of the factors with an obvious and specific macroeconomic variable such as "inflation." In fact, the latter relationship will be the subject of a heuristic regression below.

A key benefit of PCA is that it links all those criteria (separately for each of the separate factors). PCA factors are not only mean reverting but often also have a meaningful economic interpretation. Hence, they do not only fulfill the desired statistical properties but also reveal (in fundamental terms) *why* they have those properties. For example, knowing that the first factor is linked to GDP growth explains its mean reversion by the business cycles and its slow speed of mean reversion by the length of those cycles.

Hence, PCA decomposes the market into uncorrelated mean-reverting factors, which often carry an economic meaning. In other words, PCA allows us to combine statistical analysis with economic analysis so as to associate statistical features of a trade with particular fundamental considerations. For example, the second factor may be negatively correlated to the EUR exchange rate. In this case, a statistically attractive steepening trade looks even more appealing to an investor if he expects the EUR to weaken for fundamental reasons.

While statistical analysis is by construction backward-looking, linking it to fundamental variables enables traders to incorporate forward-looking (economic) expectations. This ability to incorporate potential future risks in the analysis is a key benefit of linking statistics to external driving forces. In the example above, a steepening trade may well look statistically attractive, but this could all be due to EUR strength. Thus, a further EUR strengthening, which may be caused by political decisions independent of any statistical properties, is a risk to the trade. The link of statistics through a PCA to these external driving forces can identify these risks. An analyst knowing that the statistically attractive steepness is due to EUR strength will refrain from the trade if he sees a significant risk for further EUR appreciation. As in this example, outside information about macroeconomic events can help with both explaining some observed statistical properties and incorporating forward-looking information into the analysis.

The general form of this analysis is to investigate the following link, (e.g. via a regression):

$$\text{Factor} \sim \text{External explaining variable(s)}$$

While by construction this type of analysis is outside of the statistical reach of PCA, PCA both *enables* this analysis by generating the dependent variable and *facilitates* the search for relevant external explaining variables by revealing the meaning of its factors. One could investigate the link to external variables by heuristic methods, for example, by trying all available financial time series in a regression table against all factors. Note that some financial time series could be trending and therefore not suitable for a regression.[11] The interpretation of

[11] Over the time period used for Table 3.1, the independent variables – including the S&P500, EUR, and Oil – did not exhibit a significant trend and are therefore included. Of course, this could be different over other time periods.

TABLE 3.1 Correlations of the First Three Factors of a PCA on the Bund Yield Curve versus Candidates for External Explaining Variables

	Factor 1	Factor 2	Factor 3
Factor 1 of a PCA on USD swaps	0.73	0.62	−0.10
Factor 2 of a PCA on USD swaps	0.89	0.40	−0.02
Factor 3 of a PCA on USD swaps	−0.21	−0.59	0.31
5Y Bund vol (6M rolling)	−0.52	−0.65	0.03
S&P500	0.60	−0.55	0.11
VIX	−0.72	0.01	−0.16
EUR FX rate	0.44	−0.63	−0.23
Oil	0.58	−0.63	0.08

the eigenvectors can facilitate the search for the "right" explanatory variables. As these relationships evolve, we recommend monitoring them over time, for example, via rolling correlations. For the example of a PCA on the Bund yield curve, the correlations versus some candidates for external driving forces[12] are summarized in Table 3.1.

It can be seen that both factor 1 and factor 2 are significantly influenced by a number of macroeconomic variables. We note in particular the strong link to the US yield curve (as represented by the factors of a PCA) with the surprising fact that factor 1 of the Bund PCA is most correlated to factor 2 of the USD PCA and factor 2 of the Bund PCA to factor 1 of the USD PCA. This could be the starting point for a further investigation, which may reveal interesting differences in the driving forces of global bond markets and their interconnections. Furthermore, the link between factor 2 and currencies as well as commodities jumps out at you, indicating that further analysis may well yield valuable results. Factor 3, on the other hand, seems to be rather uncorrelated to external driving forces. This could indicate that factor 3 is a relatively "pure" relative value factor (i.e. with little correlation to macroeconomic events). This is a typical result, that is, while the lower factors often exhibit a high correlation to macroeconomic variables (reflecting the high impact of economic events on markets), higher factors are usually less correlated. As a rule of thumb, the higher the factor, the more weight statistical analysis carries, while the examination of potential external economic risks and the incorporation of forward-looking analysis described above become less important.

[12]Additionally, economic variables like inflation or GDP (growth) could be included. However, given the low frequency of these data (e.g. quarterly), it is statistically meaningful only for time series spanning several years, not in the current example, which has less than two years of data.

As an alternative to the regression of PCA factors versus explaining variables, which is done outside of the PCA itself, one could also include the candidates for explaining variables in the PCA, for example, by running a PCA on input data consisting of time series for Bund yields and USD swap rates simultaneously.

Furthermore, PCA may reveal how flows affect the pricing. For example, how much does the *k*-factor residual for 5Y rates move when a new 5Y bond is issued? Is this a stable pattern? Is the spike linearly dependent on the issuance size? This allows traders to incorporate more technical issues into the framework of a PCA.

Let's see how the ability of PCA to decompose the Bund yield curve into economically meaningful uncorrelated and statistically mean-reverting factors could support finding trade ideas in practice. Figure 3.13 shows the evolution of the first three factors of the Bund yield curve over time.

Using mean reversion models, we can assess for all factors the distance from their long-run means and the speed with which they are likely to return to these means. In our current example, we may conclude that factor 1 has insufficient speed of mean reversion and thus we take no view. Similarly, factor 2 is relatively close to its mean, so we take no view. Factor 3, however, seems to be considerably away from its mean and to have a high speed of mean reversion. Hence, we decide to investigate further for trade ideas based on factor 3 (i.e. butterflies hedged against factors 1 and 2).

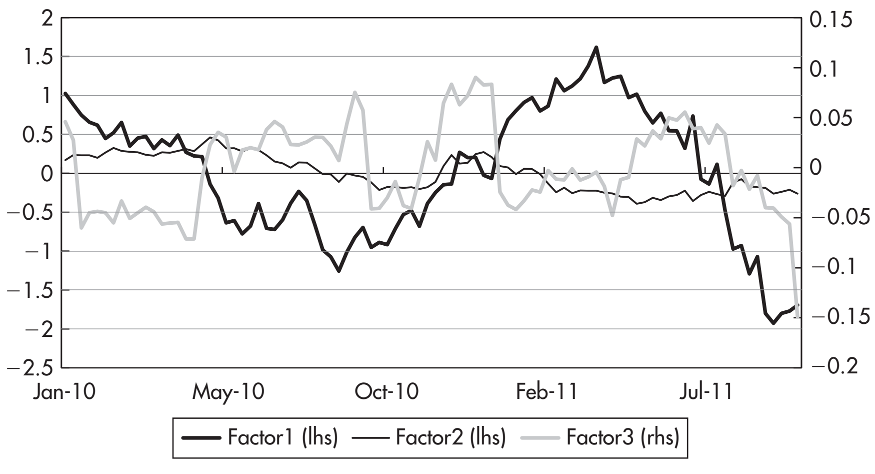

FIGURE 3.13 History of the first three factors of a PCA on the Bund yield curve.
Sources: data – Bloomberg; chart – Authors.

Data period: 4 Jan 2010 to 3 Oct 2011, weekly data.

The statistical side of that investigation is done by a formal mean reversion model, which can calculate the expected return and risk attributes of various trades for specific time horizons.

Factor 3 has little exposure to economic variables and can thus be treated as a "pure" relative value play on mean reversion, with little risk of its statistical properties being thrown off track by macroeconomic events. In this case, the regression Table 3.1 can serve as justification for restricting the analysis to statistics.

If we took a view on factor 1, on the other hand, we would need to consider the link of the time series of factor 1 and its statistical properties to its external driving forces. Since we found factor 1 to be positively correlated to USD rates and negatively to volatility, among others, the spike in the VIX on 3 Oct 2011 could help us understand the current statistical deviation from the mean. And if we thought that volatility was going to decrease again soon (e.g. because we expected the ECB to calm fears about the euro crisis), we would have both a statistical (mean reversion) and a fundamental reason to bet on an increase in factor 1. Conversely, the mean reversion of factor 1 (as given by statistics) certainly supports our fundamental expectation for volatility to decrease again (i.e. for the current point to be an outlier, i.e. a good entry opportunity). Additionally, one could run a regression between the VIX and factor 1 and decide via the current residual whether the fundamental expectation of decreasing volatility would be better expressed by a short option or a short Bund position.

As in this example, PCA guides the analyst through the process of taking informed views (or the informed absence of a view) on each of the uncorrelated factors: it reveals the statistical properties and economic meaning of each factor behind a position and gives thereby a good basis for a reasonable and substantiated decision, separately for each factor, whether or not to take a view on it.

Once the analyst has decided on which factors he wants to take a view, PCA does both jobs that are needed to execute that view via the most suitable position:

- PCA provides the hedge ratios to immunize against exposure to factors to which he does not want to be exposed.
- The $(n-1)$-factor residuals indicate those individual instruments that provide the best execution for his view on factor n. This can also be used to apply PCA for general asset selection purposes.

In the following, we will develop these features of PCA, before we integrate them back into the process of finding and analyzing trades.